

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks: 40

Annexure A

CASE STUDY

Code of Conduct:

I, Tharinda (192412004) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tharinda

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Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	24
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	23
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	18
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	18
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	8

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MP

1. Vehicle Fault Diagnosis System

2. Propositional Logic

Let :

* E = Engine Warning Light

* N = Unusual Noise

* O = Low oil Pressure

* H = Overheating Sign

* F = Engine Failure Risk

* L = Oil Leak

* C = Coolant System Failure

Rules :

$$R_1 : E \wedge N \rightarrow F$$

$$R_2 : E \wedge O \rightarrow L$$

$$R_3 : N \wedge H \rightarrow C$$

Facts :

~~E = True~~ , N = True , H = True

6. Forward Chaining

Step 1: E and N are True

Using R₁:

$$E \wedge N \rightarrow F$$

Therefore :

$$F = \text{True}$$

step2: R1 needs E and O, but O is not given. Hence no oil leak is derived.

step3: N and H are true

using R3:

$$N \wedge H \rightarrow C$$

Therefore:

$$C = \text{True}$$

Final conclusions:

Engine Failure Risk, Coolant System Failure

2 Backward chaining

Goal:

$$C = \text{Coolant System Failure}$$

R3 gives:

$$N \wedge H \rightarrow C$$

So the goal is reduced to:

$$C \Leftarrow N \wedge H$$

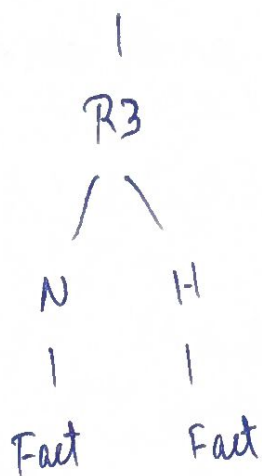
Both N and H are given facts.

Therefore:

$$C = \text{True}$$

Goal Tree

Coolant System Failure



22 Hospital Emergency Word Routing

2 Unification

Rule 1:

$\text{Patient}(x) \wedge \text{Critical Condition}(x) \rightarrow \text{Priority Corridor}(x)$

Facts:

Patient (Patient A)

Critical Condition (Patient A)

Matching gives:

~~Patient(x) = Patient (Patient A)~~

Therefore:

$\theta = \{x/\text{Patient A}\}$

Hence:

Priority Corridor (Patient A)

6) Resolution Proof

Rule 1 in CNF:

$$\neg \text{Patient}(x) \vee \neg \text{Critical Condition}(x) \vee \text{Priority Corridor}(x)$$

Using the facts:

Patient (Patient A), Critical Condition (Patient A)

We derive:

Priority Corridor (Patient A)

Rule 2:

$$\neg \text{Priority Corridor}(x) \vee \text{Corridor}(\text{Patient A})$$

Therefore:

Corridor cleared (Patient A)

Rule 3 in CNF:

$$\neg \text{Patient}(x) \vee \neg \text{Waiting Behind}(x, y) \vee \neg \text{Corridor cleared}(y) \vee \text{Hold}(x)$$

Substitute:

$$x = \text{Patient B}, y = \text{Patient A}$$

Using,

Patient (Patient B)

Waiting Behind (Patient B, Patient A)

We get:

Corridor cleared (Patient A)

Resolution chain:

Patient A + Critical Condition (A)



Priority Corridor (A)



Corridor cleared (A)



Patient B + Waiting Behind (B, A)



Hold (Patient B)

Two Critical Patients

Rule 1 applies to every Patient :-

$\forall x$

So two critical Patients can both obtain:

Priority Corridor (Patient A)

and

Priority Corridor (Patient B)

However, the KB does not specify which Patient gets priority first.

An additional rule such as:

More critical (x, y) $\rightarrow$ Priority First (x, y)

is required for conflict resolution.

Q3) Weather - Based Flight Scheduling

Q) CNF conversion

Let S = Storm Warning, D = Flight Delay Needed, I = International Route, N = Notch Passenger.

C = Compliance Risk.

P1:

$$S \rightarrow D$$

$$\neg S \vee D$$

P2:

$$D \wedge I \rightarrow N$$

$$\neg D \vee \neg I \vee N$$

P3:

$$\neg N \wedge I \rightarrow C$$

$$N \vee \neg I \vee \neg C$$

Facts:



Q) Resolution Refutation

Goal:

N

Negate goal:

$\neg N$

From

S

and:

$$\neg D \vee \neg I \vee N$$

With D :

$$\neg I \vee N$$

Using I :

$$N$$

But we have

$$\neg N$$

Therefore :

$$N \wedge \neg N \Rightarrow \boxed{\text{Q}}$$

Hence :

(Notify Passenger is proved)

2 Removing Storm Warning

Without S :

$$D$$

also cannot be derived

For Compliance Risk:

$$\neg N \wedge I \rightarrow C$$

Although I is true, $\neg N$ is not given.

Therefore

Compliance Risk cannot be concluded

24/ Cave Exploration Knowledge Agent

c) Belief state

Given:

Draft [1, 3], Glow [1, 1], sparkle [2, 3]

Using the rules:

Draft [1, 3] $\rightarrow$ Pit Adjacent [1, 3]

Therefore:

Pit Adjacent [1, 3]

Glow [1, 1] $\rightarrow$ Lava Vent Adjacent [1, 1]

Therefore:

Lava Vent Adjacent [1, 1]

sparkle [2, 3] $\rightarrow$ Treasure [2, 3]

Treasure [2, 3]

b) Forward Chaining

From Draft [1, 3]:

Draft [1, 3] $\rightarrow$ Pit Adjacent [1, 3]

Possible Pit locations:

[1, 2], [1, 4], [2, 3]

From Glow [1, 1],

Glow [1, 1] $\rightarrow$ Lava Vent Adjacent [1, 1]

From sparkle [2,3]:

Sparkle [2,3] $\rightarrow$ Treasure [2,3]

c) Path safety:

Path:

[1,1] $\rightarrow$ [1,3] $\rightarrow$ [2,3]

Rule for safety:

$\neg \text{Pit}[x,y] \wedge \neg \text{Lawyer}[x,y] \rightarrow \text{Safe}[x,y]$

But the KB does not provide negative facts proving that these cells have no Pit or Lawyer.

Also, [2,3] is a possible Pit location.

Therefore, the Path cannot be guaranteed safe

(Path is Not confirmed safe)

That agent needs additional information before collecting the treasure